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Pin-holes Based Description of the PANCAM, a Hyper-hemispherical Lens

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Summary: Hyper-hemispherical lenses are optical objectives characterized by an ultra-wide field of view. Due to their extreme field of view, geometric calibration poses significant configuration challenges. Various methods have been developed to simplify the geometric calibration of cameras in short-range fields in controlled environments, such as clean rooms, and the use of simple calibration checkerboards. The lens analyzed in this study (**PANCAM**) is intended to be a planetary payload for an ESA mission with photogrammetric objectives. Consequently, the current state-of-the-art for this type of lens has been reviewed and extended to improve the descriptive model for the hyper-hemispherical field of view projection. While the resulting dewarping is essential for photogrammetric applications, it became necessary to define an approximate model to represent a single capture as a composite of multiple pinhole models. Each of these models can be individually leveraged as a data source for scientific analyses relying on this specific geometric framework.

Keywords: Hyper-hemispherical, Panoramic, Dewarping, Calibration.

1. Introduction

Panoramic imaging can be achieved through three main methods: (i) moving cameras that stitch multiple images together, (ii) multi-camera systems with independent lenses, and (iii) omnidirectional motionless cameras [1]. While the first two methods can be time-consuming, costly, and complex, omnidirectional cameras overcome these issues by using a single detector with no moving parts, offering benefits like low power consumption, reduced weight, and fewer failure points. Despite their lower spatial resolution, they are ideal for applications like robotics, collision avoidance and scene monitoring.

Recently, we developed two innovative lenses: the **Hyperhemispheric Lens (HHL)** [2], which captures a 360°x270° field of view (FoV), and the **Bifocal Panoramic Lens (BPL)**, which provides both a 360°x100° panoramic view and a high-resolution 20° circular view [3]. These lenses, while offering moderate resolution, have gained interest in space applications due to their motionless design, low power requirements, and lightweight construction.

1.1. Design

Both lenses feature three optical components: a panoramic catadioptric, a nested frontal optics, and a shared three-lens imaging objective.

The lenses use a shared optical layout (Fig. 1), consisting of fore-optics (catadioptric and speed-down

lens) before the aperture stop (AS) and an objective lens (OBJ) to project the image onto the focal plane.

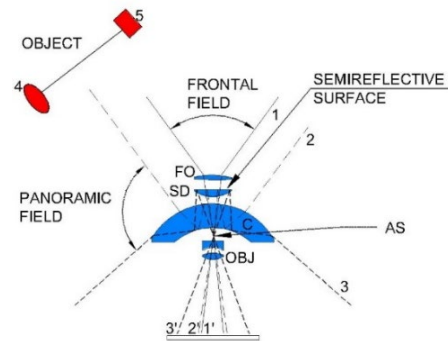


Fig. 1. HHL and BPL lenses design.

The HHL captures a 360°x270° FoV, using a catadioptric with a reflective concave surface and a semi-reflective speed-down (SD) lens to decelerate light entering the OBJ.

The BPL adapts this design by introducing a higher focal length frontal lens (FO) to enhance resolution in the central 20° circular view while retaining the panoramic FoV of 360°x100°. Both systems share a common panoramic "parent" lens, enabling easy switching between HHL and BPL configurations by replacing the FO element.

This modular design makes HHL and BPL versatile for dynamic and static scenes, offering robust solutions for panoramic imaging in space and other challenging environments.

2. PANCAM

The **PANCAM**, leveraging the BPL, combines panoramic (2 mm focal length) and scientific (3× magnification) imaging into a compact, lightweight sensor suitable for space applications (Fig. 2). More details are in Table 1.

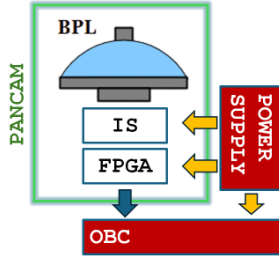


Fig. 2. PANCAM design.

Table 1. PANCAM main components.

Image Sensor (IS):	Based on the Sony IMX264 CMOS matrix with a COTS (70 krad), reliable for lunar environments
FPGA Auto-Exposure Optimization	FPGA processes and merges multiple exposures to the onboard computer (OBC) avoiding over- or underexposed areas.
Onboard Computer (OBC):	An On Board Computer (Nvidia Orin) for managing data processing and transmission.

2.1. Applications

Here are the main applications of the PANCAM, spanning from Earth's satellite environment to the lunar applications detailed in the following section: POLIFEMO [5] (A star tracker for micro-satellites and CubeSats), EMM-PANCAM (the Earth-Moon-Mars project, for planetary surface monitoring), PANVISION [8] (deep-learning-based surveillance for Covid-19 emergency) and DaedalusCAM. The last one is a payload selected in 2021, for ESA-funded DAEDALUS (Descent and Exploration in Deep Autonomy of Lava Underground Structures) and currently the Italian National Space Agency (ASI) is funding the enhancement of Daedalus CAM, to explore and increase its immersive camera capabilities with a focus on the software for displaying the immersive image of a lunar cave.

3. Daedalus Mission

The exploration of lunar subsurface lava tubes, accessible through skylights, has gained interest due to an ESA initiative under the SysNova Lunar Caves studies. The selected target is a skylight in the volcanic Marius Hills area of Oceanus Procellarum (303.3°E,

14.2°N), thought to lead to a vast underground void. Lava tubes, as highlighted by past missions like Kaguya, LRO, and GRAIL, offer significant scientific and exploration potential: they are natural shields (from micrometeorites, cosmic rays and radiation), they have stable environments and give resource access (i.e. Potential water sources). These advantages could redefine lunar exploration and pave the way for future Martian missions.

The DAEDALUS robotic sphere is designed for exploring lunar lava tubes. The payload includes four PANCAM units, enabling immersive stereoscopic monitoring during cave descent; in this phase Laser scanners and PANCAM panoramic cameras will generate a 3D point cloud of the skylight.

The scientific objectives are detailed in [6], and payload specifications in [7].

4. Calibration and Equirectangular Products

This work aims to better understand the panoramic field of view (FoV) of the PANCAM. For this reason, the Frontal Field was not mounted during this phase. An example of an image acquired in Padua Institute is shown in Fig. 3.



Fig. 3. Example of the panoramic FoV of the PANCAM.

The PANCAM is able to acquire images below the horizon making classical methods for calibration of more common cameras unsuitable.

Computer Vision researchers have affined, in the last decades, methods [9] for the calibration of simple pinhole cameras by the use of non-controlled acquisition of a known pattern (generally a chessboard). In the case of wider camera (more definable as panoramic), the best generic model for the description of an ultra-wide angle optical system is:

$$X = D \begin{bmatrix} \sin \alpha & \cos \theta \\ \cos \alpha & \cos \theta \\ \sin \theta \end{bmatrix} \rightarrow \bar{x} = \rho \begin{bmatrix} \sin \alpha \\ \cos \alpha \end{bmatrix} \quad (1)$$

The equation associates the 3D point X with the 2D point $\bar{x}$ described by the azimuthal angle and ρ the central point of the sensor.

This map, as performed by the well-known calibration tool designed by Scaramuzza [10], can be generically defined by a polynomial function f :

$$\tan(\theta) = \frac{f(\rho)}{\rho}, \quad (2)$$

where ρ is the pixel distance from the boresight and θ is the equatorial angle.

Various historical approximations have been used to define the f function: a notable calibration approach for panoramic cameras was presented in [9].

However, this central-focused methodology is not well-suited to the extreme field of view (FoV) of the PANCAM. This limitation led us to develop a more adaptable A-Central Model Extension (AME) [4] which extends Eq. (2) by incorporating two different polynomial definitions for the narrower and outer regions of the FoV while accounting for the system's acentricity.

This AME model provides a seamless transition between the simpler central projection and the more precise non-central projection, so accommodating a wide range of fisheye lens designs. Furthermore, it extends the ability to model wide-angle cameras, including hyper-hemispherical lenses, which previous tools have not adequately addressed.

The product of the AME is a direct map we can define as

$$\Omega_{\text{AME}}: Az, Ze \rightarrow x, y \quad (3)$$

As reported in [4] the AME model achieves a reduction of more than 25 % in the mean residual error compared to Scaramuzza's approach.

As reported in [4] if classical approaches as Scaramuzza can reach a standard deviation of 0.62/0.57 (x/y) pixel in the residual of the extracted corner and an average of 0.82 px, AME model reach a std of 0.5/0.4 px with an average of 0.6 px. Aside from these statistical data, it should be emphasized (as

shown in the maps presented in [4]) that the distribution of these errors in the AME is completely uniform and not related, as in classical models, to the position with respect to the edge of the field of view.

5. PANCAM Products Reduction

Fig. 4 shows the equirectangular projection of the image shown in Fig. 3 based on AME Model [4].

We have to consider that the DAEDALUS mission has selected PANCAMs as panoramic cameras due to their hyper-hemispheric configuration, which ensures a complete view of the pit and provides extensive redundancy. On the other side the onboard Lidars will generate a point cloud of the pit, enabling the projection of PANCAM images to create orthorectified visuals.

However, a part the case of a detailed surface reconstruction, dewarping PANCAM images remains challenging. The AME model accurately predicts the camera's geometry but requires a calibration campaign based on acquiring dozens of images of a calibration pattern, followed by subsequent post-processing.

Moreover, while the AME model has already been applied and equirectangular projection is currently possible, as shown in Fig. 4.

In this projection, lens anamorphism adds to the inherent distortion of the projection itself. The projection introduces a polar distortion—similar to what we observe in Earth maps. Additionally, if the image is almost diffraction limited (with an angular resolution of 0.1°/px) from the azimuth down to the horizon. For angle below the horizon, the image quality deteriorates due to the anamorphism: as the zenith angle increases across the ultra-wide field of view, the instantaneous field of view (IFOV) reaches up to 5 mrad, introducing aberrations and distortions that are difficult to correct using standard calibration pipelines.



Fig. 4. Equirectangular projection of image Fig. 3.

Equirectangular images (as the one shown in Fig. 4) are based on the chief rays map measured thanks to AME model (see Fig. 5). Despite their angular and geometric accuracy, the change of anamorphism introduced in this projection and the distortion (described in the last paragraph) make these images difficult to process using traditional imaging

methods, which are typically designed for pinhole camera systems. This is illustrated in Fig. 5 by the noticeable misalignment between the zenithal curves (in red) and the vertical chessboard panel. The zenithal curves align perfectly with the row of corners near the equatorial FoV but gradually transform into maximal circles in the region near the boresight.

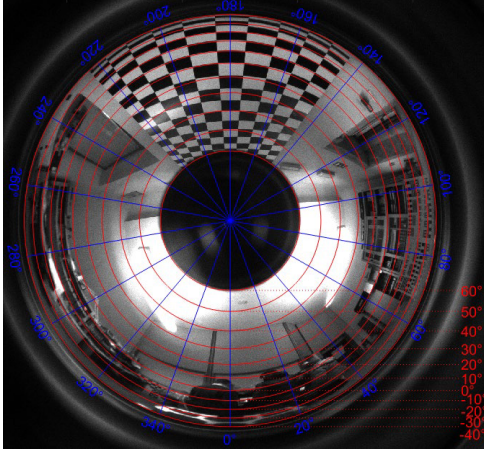


Fig. 5. Azimuth (in blue) and Zenith (in red) mapping measured by AME on a PANCAM image.

Moreover, anamorphism and distortion do not preserve homothetic information, causing all straight lines to be transformed into conic curves.

5.1. Pinholes Approach

On the other hand, more common tools for image analysis are based on pinhole geometry. For this reason, we investigate the possible approximations of the map Ω_{AME} described in Eq. (3) with a simpler description. Tests have underlined that polynomial functions of degree 5 in terms of the Azimuth (A_z) and Zenith (Z_e) angle can be considered as local approximations of the x, y pixel coordinates:

$$x/y = P_{x/y}(A_z, Z_e) = \sum_{i=0}^5 c_i A_z^i Z_e^{5-i} \quad (4)$$

These parameters can be defined by a unique acquisition of a well-known target covering all the Zenithal FoV and a part of the Azimuth FoV as shown in Fig. 5. From an Azimuthal point of view the geometry of the lens can be extended for radiality.

Knowledge of a limited portion of the FoV (both in metric and angular terms) can provide an approximate polynomial definition that can be extended to a larger FoV. Assuming the azimuthal angle is divided into 60° segments, each section can be approximated by a polynomial defined in Eq. (2).

These parameters must be derived from a measurement (on an image) that covers a smaller FoV.

Fig. 6 illustrates three tests identifying regions near the boresight (in red), at a mid-range zenith angle (in green), and close to the hyper-hemispheric region.

For each of these regions, progressively larger FoVs (ranging from 10° to 60°) were considered. Using the information contained in the image, polynomials were defined to approximate the mapping.

Approximation requires more FoV in the case of the region near boresight where, as shown in Fig. 5 anamorphisms change faster. In any case, for all the

regions a FoV of 50 degrees is enough to reach a sub-pixel approximation.

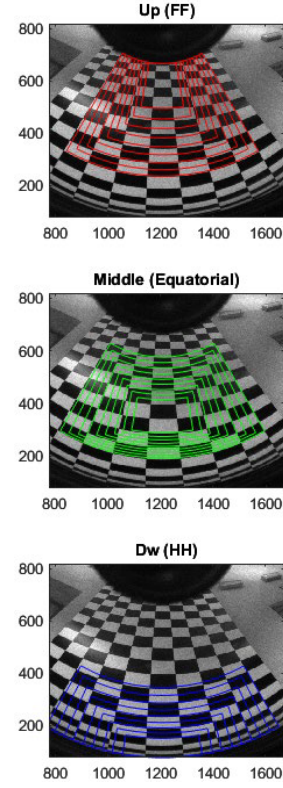


Fig. 6. Different regions of the FoV used for the definition of the polynomial approximation.

Fig. 7 shows the standard deviation of the residuals (in horizontal and vertical direction) between polynomial approximation and map measured by AME as function of the dimension of region of FoV used as measuring region in the three cases (reported with the same color) shown in Fig. 6.

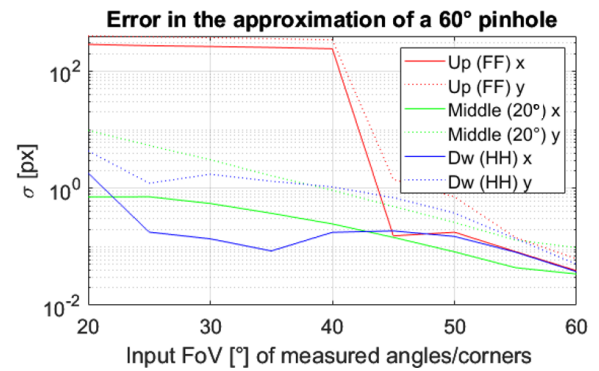


Fig. 7. The residuals between polynomial approximation and map measured by AME.

Defined a versor v with a zenith angle, coherent with the region considered, it is possible to use v as a boresight of a virtual pinhole camera with a fixed IFOV ϵ (we assume the IFOV as the minimum one of the PANCAM lens which means 0.1°).

$$\begin{cases} v = [0, \sin(Z_e), \cos(Z_e)] \\ v_x = [0, 0, 1] \\ v_z = v_x \times v \end{cases}$$

Following this definition each chief ray $\tilde{v}_{ij}$ of the pinhole camera can be defined as:

$$\tilde{v}_{ij} = v + i\epsilon v_x + j\epsilon v_y$$

$$\begin{cases} A_{zij} = \text{atan} \left(\frac{\tilde{v}_{ij}(1)}{\tilde{v}_{ij}(2)} \right) \\ Z_{eij} = \text{atan} \left(\frac{\tilde{v}_{ij}(3)}{[\tilde{v}_{ij}(1) \tilde{v}_{ij}(2)]} \right) \end{cases}$$

To each of the chief ray is associated the angular couple:

Each pixel of the virtual camera can be associated to each pixel of the original image via bilinear interpolation.

4. Conclusions

The pinhole approach to PANCAM images allows for describing the hyper-hemispherical camera in a

more direct way, making geometric data more immediately accessible to well-known algorithms in the photogrammetric field and avoiding the development of complex methodologies to address the lens anamorphism and issues related to different types of projections.

The result of this process can be seen in Fig. 8 where the virtual pinholes are shown next to the equirectangular projection. Note that the architectural features highlighted in yellow and orange squares maintain their Euclidean projection in the pinhole versions. Moreover, the chessboard pattern (blue and green), which is significantly affected by the anamorphism introduced by both the lens and the equirectangular projection, is well represented in the approximated version as simple homographies. These examples illustrate the compatibility of this approach with most image analysis methods, making the output of the PANCAM more easily usable for scientific and engineering applications.

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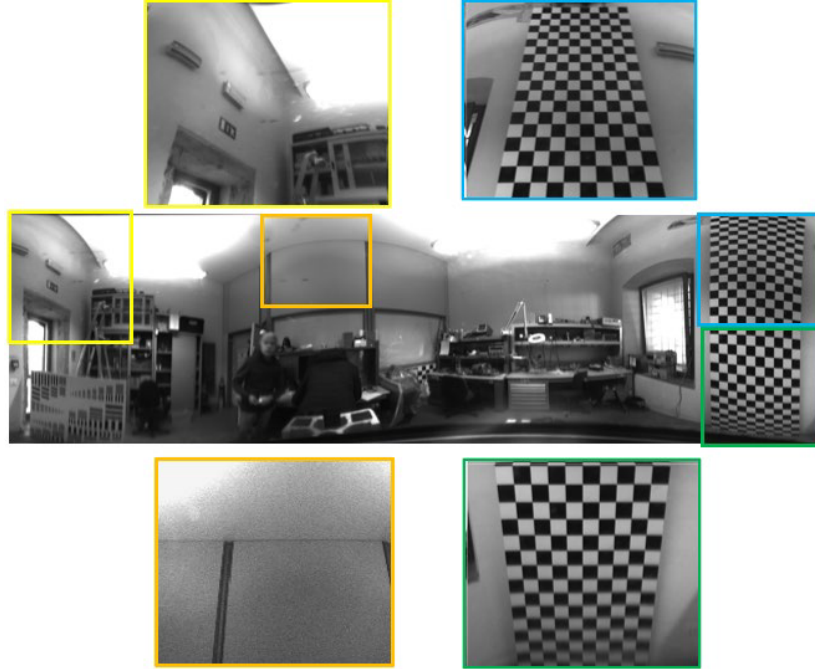


Fig. 8. Comparison between the equirectangular dewarping and a sample of a virtual pinhole quickly generated by the polynomial approximation.

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